ALGEBRA

AMIRREZA ADHAM

Algebra

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Contents

1 The Whole Numbers

1.1 Place Value, Names for Numbers, and Tables

The **whole numbers** are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, ...

The **natural numbers** are 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, ...

Finding the Place Value of a Digit in a Whole Number

The position of each digit in a number determines its **place value**.

Periods											
Billions			Millions			Thousands			Ones		
Hundred-billions	Ten-billions	**Billion**	Hundred-millions	Ten-millions	**Millions**	Hundred-thousands	Ten-thousands	**Thousands**	Hundreds	Tens	**Ones**

Writing a Whole Number in Words and in Standard Form

A whole number such as 1,083,664,500 is written in **standard form**. Commas separate the digits into groups of three, starting from the right. Each group of three digits is called a **period**.

Writing a Whole Number in Words

To write a whole number in words, write the number in each period followed by the name of the period. This same procedure can be used to read a whole number.

Example

9, 265 is read as "nine thousand, two hundred sixty five."

We write 1, 083, 664, 500 as "one billion, eighty-three million, six hundred sixty-four thousand, five hundred".

Writing a Whole Number in Standard Form

To write a whole number in standard form, write the number in each period, followed by a comma.

Writing a Whole Number in Expanded Form

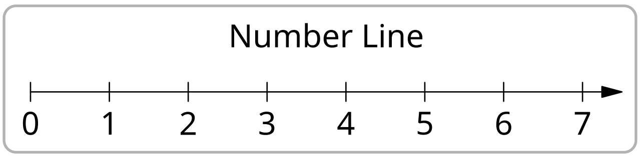
The place value of a digit can be used to write a number in expanded form. The **expanded form** of a number shows each digit of the number with its place value.

Example

5672 is written in expanded form as

$$5672 = 5000 + 600 + 70 + 2$$

We can visualize whole numbers by points on a line. The line below is called a **number line**. This number line has equally spaced marks for each whole number. The arrow to the right means that the whole numbers continue indefinitely.



Tables

Tables are often used to organize and display facts that contain numbers.

Number of notebooks	Cost per notebook	Total cost
1	$3	$3
2	$3	$6
3	$3	$9

Number of notebooks	Cost per notebook	Total cost
4	$3	$12
5	$3	$15

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